ELECTRIC VEHICLE MARKET PERFORMANCE AND INSIGHTS
(EV MARKET TRAIN) — PROJECT REPORT

1. INTRODUCTION

1.1 Project Overview

The **Electric Vehicle Market Performance and Insights** project focuses on analyzing the global electric vehicle (EV) industry using data visualization and business intelligence techniques in Tableau.

The electric vehicle industry is growing rapidly due to increasing environmental awareness, government initiatives, technological advancements, and the demand for sustainable transportation.

This project uses the `ev_market_train.csv` dataset, containing comprehensive information about electric vehicle manufacturers, models, battery specifications, vehicle categories, manufacturing countries, market regions, prices, driving range, annual sales, customer ratings, and environmental impact. The project aims to transform raw EV market data into meaningful visual insights through interactive Tableau dashboards.

1.2 Purpose

The main purposes of this project are:

- To analyze global electric vehicle market performance.
- To compare manufacturers based on annual sales and market share.
- To study different vehicle categories and battery technologies.
- To analyze vehicle prices, battery capacity, and driving range relationship.
- To understand customer ratings and consumer satisfaction levels.
- To examine the environmental benefits and CO_2 reduction of electric vehicles.
- To provide useful insights for business decisions and sustainability research.

2. IDEATION PHASE

2.1 Problem Statement

The global electric vehicle market generates a large amount of data related to manufacturers, vehicle specifications, battery technologies, sales, pricing, and environmental performance. However, understanding this information through raw datasets is difficult and time-consuming.

There is a need for an interactive data visualization solution that can organize and present EV market information in a simple and understandable format.

> **Problem Statement: To develop an interactive Tableau-based dashboard that analyzes global electric vehicle market performance and provides meaningful insights into sales, manufacturers, battery technologies, vehicle categories, pricing, customer satisfaction, and sustainability.**

2.2 Empathy Map Canvas

Target Users: (1) Emily Johnson – Market Strategy Analyst, (2) David Chen – Business Intelligence Analyst, (3) Dr. Sarah Williams – Sustainability Researcher.

Says	Thinks	Does	Feels
• Need global EV trends. • Want to compare OEMs. • Reliable sustainability data.	• Which OEM performs best? • Which battery is popular? • How price impacts range?	• Analyzes sales data. • Compares specs. • Studies environmental impact.	• Needs accurate info. • Wants interactive UI. • Wants data-driven decisions.

2.3 Brainstorming

Key ideas considered during development include global EV sales analysis, manufacturer sales comparison, vehicle category distribution, battery technology analysis, price vs. driving range, customer ratings, and CO_2 savings analysis. These ideas were organized into three dashboard scenarios: **Market Strategy**, **Business Intelligence**, and **Sustainability Research**.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Activity	Expected Outcome
Awareness	Opens the EV dashboard	Understands project purpose
Exploration	Selects filters and categories	Explores EV market data
Analysis	Compares sales and specifications	Identifies useful patterns
Evaluation	Studies ratings and sustainability	Evaluates EV performance
Decision	Reviews dashboard insights	Supports informed decisions

3.2 Solution Requirement

Functional Requirements: Import & analyze EV dataset; display global EV metrics; show manufacturer sales; compare vehicle categories; visualize battery types/capacities; analyze price & driving range; display manufacturing countries; analyze ratings & CO_2 savings; provide interactive filters.

Non-Functional Requirements: Easy-to-understand design; accurate data representation; interactive & user-friendly UI; consistent formatting; efficient rendering performance.

3.3 Data Flow Diagram

Level 0 – Context Diagram: External Data Source → EV Market Analysis System (Tableau) → Interactive Dashboards → End Users

Level 1 – Data Flow Steps: (1) Collect EV dataset → (2) Import CSV into Tableau → (3) Prepare & clean dataset → (4) Create calculated fields → (5) Develop 3 scenario dashboards → (6) Display interactive reports.

3.4 Technology Stack

Technology	Purpose
Tableau Desktop 2025/2026	Data visualization and interactive dashboard development.
CSV (ev_market_train.csv)	Dataset storage and input data source.
Microsoft Excel	Data inspection and basic pre-processing.
Computer System	Development environment and project execution.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The project addresses the difficulty of analyzing large EV market datasets by presenting critical information through interactive Tableau visualizations. Instead of examining thousands of raw records, users explore sales, manufacturers, battery specs, pricing, range, ratings, and sustainability metrics dynamically.

4.2 Proposed Solution (3 Scenario Dashboards)

Scenario 1: Market Strategy (Emily Johnson) EV sales by vehicle category, manufacturer sales comparison, customer rating analysis, battery tech comparison, battery capacity vs. driving range.

Scenario 2: Business Intelligence (David Chen) Manufacturer annual sales, market share comparison, battery type distribution, manufacturing country analysis, price vs. sales analysis.

Scenario 3: Sustainability Research (Dr. Sarah Williams) CO_2 savings by vehicle/category, battery capacity & range efficiency, manufacturing country distribution, customer satisfaction, vehicle environmental impact.

4.3 Solution Architecture

Data Source (ev_market_train.csv) ➔ Data Inspection & Cleaning ➔ Tableau Desktop ➔ Chart & Map Development ➔ 3 Scenario Dashboards ➔ Interactive Final Report

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activity
Phase 1	Project topic selection and scoping.
Phase 2	Dataset collection (ev_market_train.csv).
Phase 3	Dataset inspection, cleaning, and preparation.
Phase 4	Connecting dataset to Tableau Desktop environment.
Phase 5	Creating individual worksheets and visualizations.
Phase 6	Developing scenario-based interactive dashboards.
Phase 7	Formatting, styling, and configuring filter actions.
Phase 8	Testing and reviewing dashboard interactivity.
Phase 9	Preparing comprehensive final project report.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test Case	Expected Result
Dataset connection	CSV dataset connects cleanly to Tableau without field errors.
Chart visualization	All worksheets render correct measures and dimensions.
Manufacturer filter	Visualizations update dynamically according to OEM selection.
Category filter	Selected vehicle categories filter cleanly across the dashboard.
Year analysis	Available model years are represented correctly on time trends.
Dashboard navigation	All three scenario dashboards are accessible without lag.
Map visualization	Geographic data is displayed accurately on regional maps.
Interactive filters	Cross-filtering between charts functions as expected.

7. RESULTS

7.1 Output Screenshots

This section outlines the visual outputs from the completed Tableau workbook:

Screenshot 1: EV Market Overview
[High-level KPIs, total sales & market share summary]

Screenshot 2: Market Strategy Dashboard
[Emily Johnson scenario view & category distribution]

Screenshot 3: Business Intelligence View
[David Chen scenario view & OEM comparisons]

Screenshot 4: Sustainability Research View
[Dr. Sarah Williams scenario & CO_2 savings metrics]

8. ADVANTAGES & DISADVANTAGES

Advantages

- Provides a clear, comprehensive overview of the global EV market.
- Makes complex, multi-dimensional datasets easy to understand.
- Supports detailed manufacturer and vehicle category comparisons.
- Helps analyze battery technologies, range, and vehicle pricing efficiency.
- Provides valuable insights into customer satisfaction and ratings.
- Supports sustainability research and carbon reduction analysis.
- Interactive dashboards enable self-service data exploration.

Disadvantages

- Analysis accuracy depends heavily on the underlying dataset quality.
- Missing or inaccurate records may affect specific visualization outcomes.
- Tableau Desktop requires adequate computational hardware resources.
- Represents static historical data and does not automatically reflect live real-time market shifts.

9. CONCLUSION

The **Electric Vehicle Market Performance and Insights** project demonstrates how data visualization can be effectively used to analyze and understand the global electric vehicle industry.

Using the `ev_market_train.csv` dataset and Tableau Desktop, the project organizes complex EV metrics into intuitive interactive visualizations covering manufacturers, vehicle categories, sales performance, battery technologies, pricing, driving range, customer ratings, and environmental impact.

The three scenario-based dashboards provide tailored perspectives for market strategy, business intelligence, and sustainability research, emphasizing the value of BI tools in modern strategic decision-making.

10. FUTURE SCOPE

- Integration of live streaming EV market data feeds.
- Expanding dataset coverage to include more countries and emerging OEMs.
- Predictive EV sales forecasting using machine learning algorithms.
- Advanced battery degradation and performance prediction modeling.
- Comparison between electric vehicles and internal combustion engine (ICE) vehicles.
- Integration of charging station geographical location data.
- Cloud deployment via Tableau Server or Tableau Cloud for enterprise access.

11. APPENDIX

A. Dataset: `ev_market_train.csv` (Global Electric Vehicle Market Coverage)

B. Technologies Used: Tableau Desktop 2025/2026, CSV, Microsoft Excel

C. Source Code / Workbook: Tableau Workbook (`.twbx`) containing calculated fields and interactive actions.

D. Dataset Link: Available in project repository.

E. GitHub Repository: `https://github.com/ev-market-performance-insights`

F. Project Demo Link: `https://public.tableau.com/views/EV_Market_Insights`